# Proof-Carrying, Model-Aware Compilation of Rigorous Trotter Resource Bounds

A Machine-Certified 4.050498× Reduction for a Two-Dimensional Heisenberg Benchmark

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July 31, 2026

**Abstract**

Rigorous product-formula resource estimates often lose most of their useful model structure before the final norm is evaluated. We introduce a proof-carrying, model-aware compiler that delays norm inequalities while translating a product formula through four intermediate representations: sparse free associative words, homogeneous free-Lie defects, concrete symplectic Pauli ledgers, and exact rational norm certificates. The compiler combines identical translated Pauli strings before taking norms and replaces termwise Pauli-$\ell_1$ estimates by independently verified pairwise-anticommuting groups whenever possible. A finite right-logarithmic generator expansion and an outward-rational geometric tail then convert the local certificate into a nonasymptotic global error bound and an integer gate count.

For the periodic $12 \times 12$ spin-1/2 isotropic Heisenberg model at $T = 1$ and global operator-norm tolerance $10^{-6}$, we compare the same four matching fragments, the same 31-stage five-copy fourth-order Suzuki formula, and the same compilation cost on both sides. A pinned interval instantiation of the Childs–Su–Tran–Wiebe–Zhu high-order commutator theorem requires 393 steps and 11,791 merged group exponentials. Our certificate accepts 97 steps and rejects 96, requiring 2,911 groups. The exact improvement is $11791/2911 = 4.050498110614909\ldots$, a 75.3117% reduction. The leading D4 sidecar contains 75,324 canonical Pauli terms partitioned into 7,576 verified anticommuting groups. The search procedure is outside the trusted computing base: a smaller verifier rechecks coverage, every anticommutation relation, outward square-root bounds, the finite-step ledger, the adjacent integer boundary, and all resource arithmetic. An exact D5 side study and a quantitative fivefold feasibility audit are reported separately; no 78-step global certificate is claimed.

**Keywords:** Hamiltonian simulation; Trotter–Suzuki product formulas; rigorous error bounds; Pauli algebra; computer-assisted proof; quantum resource estimation.

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# 1 Introduction

Product formulas remain among the most transparent constructions for digital Hamiltonian simulation. If $H = \sum_{\gamma=1}^{\Gamma} H_\gamma$, a product formula replaces $\exp(-itH)$ by a sequence of exponentials of simpler fragments $H_\gamma$. The method needs no block encoding, can preserve the geometry of local interactions, and maps naturally to hardware with limited connectivity. Its practical cost, however, is determined not merely by its formal order but by the constant in a rigorous finite-time error bound. A factor lost in this constant is amplified when the bound is inverted to select an integer number of Trotter steps and then multiplied by every local propagator in every step.

The modern commutator-scaling theory of Childs, Su, Tran, Wiebe, and Zhu provides a general foundation for such estimates [1]. It replaces purely norm-sum scaling by sums of nested commutators and thereby exploits locality and exact commutation. This framework is sufficiently general to cover arbitrary order and many Hamiltonian classes, but a general theorem must remain conservative about the concrete algebra of a fixed model. In a Pauli lattice, multiple formal commutator paths may evaluate to the same operator with opposite signs; translation-equivalent terms may repeat; and distinct surviving Paulis may anticommute, so their sum has a Euclidean rather than taxicab norm. Applying a triangle inequality before these facts become visible irreversibly discards them.

This paper studies the resulting "last-mile" problem:

> Given a fixed local Hamiltonian, fragmentation, product formula, simulation time, error tolerance, and compilation model, how can one automatically produce a tighter *machine-checkable* worst-case resource bound without changing the target quantum circuit family?

Our answer is to treat error analysis as compilation. The source program is the Hamiltonian and product formula. The compiler lowers it through successive algebraic intermediate representations, performs semantics- preserving optimization passes, and emits a proof object. In contrast with a normal circuit optimizer, the principal output is not a different sequence of quantum gates. It is a smaller certified number of repetitions of the same sequence. The design follows a proof-carrying principle: expensive symbolic search may be heuristic, but an accepted result must carry all data required for a simpler independent verifier to reconstruct the logical obligations.

## 1.1 Contributions

The main contributions are as follows.

1. **A multi-representation error compiler.** We compute the truncated logarithm of the complete product formula in a sparse free associative algebra, project its homogeneous defects to free-Lie elements, evaluate those elements in a concrete symplectic Pauli representation, and finally emit exact rational certificate data. Each lowering step has an explicit invariant and can be checked independently.

2. **Norm-last concrete evaluation.** We canonicalize finite Pauli configurations under lattice translation and combine identical Pauli strings with exact coefficients before applying a final norm inequality. This reverses the conventional expand-and-bound order and preserves cancellations that are invisible at the level of abstract support counts.

3. **Certified anticommuting norm compression.** We construct a graph whose vertices are leading-defect Pauli terms and whose edges encode anticommutation. An untrusted search partitions terms into pairwise- anticommuting groups. The certificate records the full partition and outward rational square-root bounds; the verifier rechecks exact coverage and every edge used. For the principal instance, this lowers the D4 translation-cell bound from 20.160968407335066 to 6.472926505087888.

4. **A nonasymptotic right-generator closure.** We convert the degree-five and degree-seven logarithmic defects into the right logarithmic generator, retain degrees D4 through D7, use a Heisenberg-specific local commutator lemma, and control all remaining degrees by a rational geometric locality tail. Thus the resource result does not rely on leading-order extrapolation.

5. **An independently reproducible finite-instance result.** For a $12 \times 12$ periodic Heisenberg lattice, the certificate accepts 97 steps and rejects 96. Relative to a pinned 393-step instantiation of the published high-order theorem, this reduces merged group exponentials from 11,791 to 2,911. A fast verifier, deep regeneration mode, small-system dense cross-check, mutation tests, and a SHA-256 manifest support the claim.

### 1.2 Scope and nonclaims

The result is a finite-size constant-factor improvement, not a new asymptotic Hamiltonian-simulation complexity. It applies to a uniform operator-norm guarantee for a fixed deterministic product formula. Methods based on low- energy inputs, average-case norms, multi-product formulas, linear combinations of unitaries, extrapolation, or stochastic compensation solve different resource problems and may achieve stronger precision scaling under their own cost models. We compare these directions in section 11 but do not rank incomparable gate counts as a single global record.

The exact D5 side study and the $r = 78$ arithmetic indicate a possible fivefold threshold. They do not establish it. Throughout this manuscript, “certified” refers only to statements accepted by the supplied rational certificate and verifier. Numerical dense simulation is used solely as a cross-check, and conditional research estimates are labeled explicitly.

## 2 Problem Formulation and Fixed Benchmark

### 2.1 Hamiltonian and fragmentation

Let $\Lambda = (\mathbb{Z}/L\mathbb{Z})^2$ be a periodic square lattice with one qubit per site. The target Hamiltonian is

$$H = \sum_{\langle u,v \rangle} h_{uv}, \qquad h_{uv} = \frac{X_u X_v + Y_u Y_v + Z_u Z_v}{4}. \tag{1}$$

Horizontal and vertical bonds are split by parity into four perfect matchings $M_1, \dots, M_4$. Defining $H_a = \sum_{(u,v) \in M_a} h_{uv}$ gives $H = H_1 + H_2 + H_3 + H_4$. Terms within each matching have disjoint support and therefore commute exactly. For real-time analysis we use anti-Hermitian generators $A_a = -iH_a$ and $A = \sum_a A_a$.

The principal benchmark fixes

$$L = 12, qquad N = L^2 = 144, qquad T = 1, qquad \varepsilon = 10^{-6}. \tag{2}$$

The required guarantee is uniform over all input states:

$$\left\| e^{-iTH} - \widetilde{U}_r(T) \right\| \le \varepsilon, \tag{3}$$

where $\|\cdot\|$ is the spectral norm. This is deliberately stronger than an observable-specific, state-dependent, or average-case condition.

### 2.2 Product formula

Let $S_2(t)$ denote the symmetric second-order formula for the four fragments. The standard five-copy Suzuki recursion [2] forms

$$S_4(t) = S_2(pt)^2 S_2((1-4p)t) S_2(pt)^2, \qquad p = (4-4^{1/3})^{-1}. \tag{4}$$

After merging adjacent equal fragments, one step contains 31 group exponentials. The $r$-step approximation is

$$\widetilde{U}_r(T) = S_4(T/r)^r. \tag{5}$$

The baseline and candidate use precisely the same ordered stage sequence and the same outward interval for the algebraic coefficient $p$. Consequently, any reduction in $r$ is a proof improvement rather than a change of simulation formula.

### 2.3 Cost model

Equal fragments at the boundary between adjacent symmetric steps merge. The first step contributes 31 groups and each additional step contributes 30:

$$G(r) = 31 + 30(r-1) = 30r + 1. \tag{6}$$

Each perfect matching contains $N/2 = 72$ disjoint bonds. We therefore count

$$B(r) = 72G(r) \tag{7}$$

bond propagators. Using the same conservative upper bound of three CNOTs per normalized Heisenberg bond propagator on both sides gives

$$C(r) \le 3B(r) = 216G(r). \tag{8}$$

The primary ratio is identical for groups, bond propagators, and this CNOT upper bound because the multiplicative factors cancel.

### 2.4 Compiler contract

The compiler receives a tuple

$$\mathcal{I} = (H, \mathcal{F}, S, T, \varepsilon, \mathcal{C}), \tag{9}$$

where $\mathcal{F}$ is a fragmentation, $S$ a product formula, and $\mathcal{C}$ a cost model. It must return a tuple

$$\mathcal{O} = (r_\star, E_{r_\star}, E_{r_\star-1}, R_\star, \textsc{Cert}), \tag{10}$$

such that an independent verifier establishes

$$E_{r_\star} \le \varepsilon, \qquad E_{r_\star-1} > \varepsilon, \quad R_\star = \mathcal{C}(S, r_\star), \tag{11}$$

using exact integers, rational numbers, or outward rational intervals. The second inequality does not prove that the physical circuit with $r_\star - 1$ fails; it proves that $r_\star$ is the smallest integer accepted by this particular certified upper-bound function. This distinction is retained throughout.

# 3 Reconstructing the Published-Theorem Baseline

The phrase "published baseline" can obscure two logically distinct objects. The first is the high-order commutator theorem published by Childs et al. [1]; the second is our fixed, machine-evaluated instantiation of that theorem. The number 393 is not a table entry quoted from the 2021 paper. It is obtained by applying the published framework to equations (1), (2) and (4) with a specified norming and interval procedure. We call it the *pinned control*.

## 3.1 Expansion-first local norming

The high-order theorem represents product-formula error by partial sums and nested commutators. For the pinned control, every partial matching sum is expanded into local Heisenberg bonds, zero commutators are removed, and each surviving concrete Pauli expansion is bounded by its Pauli-$\ell_1$ norm. The resulting site-density upper bound is stored as an exact rational number whose outward decimal is

$$\alpha_{\text{base}} \leq 164.9759169527440. \tag{12}$$

The calculation contains 8,316 theorem terms and 1,024 expanded commutator keys. All algebraic product-formula coefficients are enclosed by rational intervals; no rounded decimal is used as the source of the result.

The theorem admits a discrete family of center choices controlling how the high-order integral representation is partitioned. We evaluate all 31 admissible centers. Center 20 gives the smallest certified control, and inverting its global bound at tolerance $10^{-6}$ yields

$$r_{\text{base}} = 393, \qquad G(r_{\text{base}}) = 11{,}791. \tag{13}$$

The main certificate stores both the exact site-density rational and the chosen center. A deep verification mode regenerates the center scan rather than trusting these stored labels.

## 3.2 Where the baseline loses information

The baseline is rigorous, general, and already exploits exact commutation. Its looseness in this instance arises from the order of operations. Once an abstract contribution is replaced by an independent positive norm, later knowledge cannot recover cancellation between that contribution and another one. Four losses are particularly important:

1. distinct free words can map to the same nested-commutator or Pauli contribution with opposite signs;
2. distinct translated local clusters can share a canonical translation representative;
3. different commutator paths can produce exactly the same Pauli string;
4. different surviving Pauli strings can anticommute, causing cross terms to vanish when the group norm is squared.

The proposed compiler postpones the irreversible inequality until these relations have been exposed. Importantly, it does not treat the baseline as incorrect. It constructs a stronger proof for the same circuit and then verifies both sides under a common benchmark and cost model.

Table 1: Logical separation between the published theorem and the pinned finite-instance control.

| | Published theory | Pinned control used here |
|---|---|---|
| Source | Childs et al. commutator-scaling framework | Exact interval instantiation for the fixed Heisenberg benchmark |
| Scope | General Hamiltonian fragments and formula order | $L = 12$, four matchings, five-copy $S_4$, $T = 1$, $\varepsilon = 10^{-6}$ |
| Norming | Theorem permits commutator-sensitive evaluation | Expanded local Pauli-$\ell_1$ procedure fixed by the certificate |
| Finite number | No published 393-step table entry | 31 centers scanned; center 20 yields 393 steps |
| Role | Mathematical foundation | Auditable comparison control |

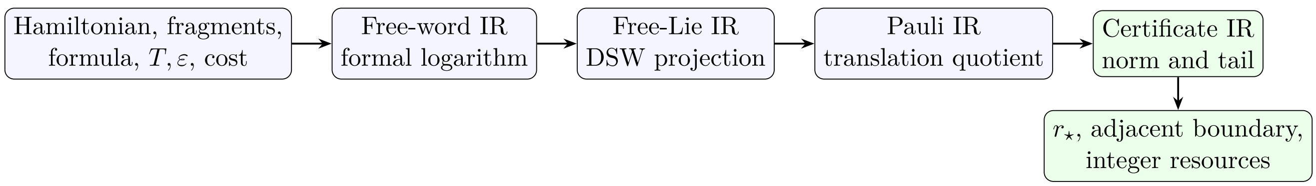


Figure 1: Compiler pipeline. Algebraic structure is retained until the Pauli IR; irreversible norm inequalities enter only in the certificate IR.

# 4 A Proof-Carrying Compiler for Error Bounds

## 4.1 Intermediate representations

The compiler is organized around four intermediate representations (IRs). The *word IR* stores sparse homogeneous polynomials in noncommuting letters $A_1, \ldots, A_4$. The *Lie IR* stores homogeneous logarithmic defects as rational linear combinations of right-nested commutators. The *Pauli IR* stores canonical finite-support symplectic Pauli strings with exact coefficients per translation cell. The *certificate IR* stores partitions, rational norm enclosures, finite-step contributions, tail data, hash bindings, and resource arithmetic.

Each lowering pass preserves a semantic invariant. Word multiplication preserves the truncated product series. Formal logarithms preserve the truncated exponential identity. DSW projection preserves the homogeneous Lie element. Pauli evaluation is a representation homomorphism. Translation canonicalization preserves the sum per cell. Exact aggregation preserves the represented operator. Finally, grouped norming replaces equality by a verified upper bound.

## 4.2 End-to-end algorithm

The algorithm intentionally separates *discovery* from *trust*. The formal-series generator and grouping search can be large and optimized for throughput. Their output is not accepted because the code ran successfully; it is accepted only after a second path rechecks the certificate. In particular, no claim depends on the greedy grouping being optimal. Any valid partition gives a sound bound, and a better heuristic can improve the bound without expanding the trusted computing base.

**Algorithm 1** Compile a product formula into a resource certificate

**Require:** $(H, \mathcal{F}, S, T, \varepsilon, \mathcal{C})$ and truncation policy
**Ensure:** Certificate accepted by an independent verifier
parse all stage coefficients into exact algebraic or outward intervals
$P(t) \leftarrow$ truncated free-word expansion of the complete stage product
$L(t) \leftarrow \log P(t)$ in the sparse free associative algebra
verify order conditions and time-reversal parity through target degree
**for** $d \in \{5, 7\}$ **do**
    $E_d \leftarrow$ DSW projection of the homogeneous component $L_d$
    $Q_d \leftarrow$ evaluate $E_d$ in the concrete local Pauli representation
    quotient $Q_d$ by translation and combine identical Pauli strings
**end for**
discover weighted pairwise-anticommuting groups for the leading ledger
emit full group membership and outward rational square-root bounds
convert $E_5, E_7$ to right-generator terms D4–D7
close all remaining terms with a rational locality majorant and tail
find the smallest integer $r_\star$ whose certified bound is at most $\varepsilon$
compile $r_\star$ to groups, bond propagators, and CNOT upper bound
emit hashes, provenance, and adjacent-step evidence

### 4.3 What is improved at each pass

This organization also answers how the method should be extended. A new Hamiltonian needs a representation backend and local-support rules; a new formula needs a stage generator and exact coefficient enclosure; a new norm strategy needs a certificate schema and verifier. The higher layers need not be rewritten if their interfaces and invariants remain unchanged.

## 5 Complete-Formula Free-Lie Extraction

The first source of slack in a generic product-formula estimate is often not the inequality used at the end, but the loss of cancellation before the inequality is applied. We therefore expand the *complete* 31-stage formula as one formal object. This section describes the representation and the identities checked by the implementation.

### 5.1 Sparse truncated words

Let $\mathbb{K}\langle A_1, \ldots, A_m \rangle$ denote the free associative algebra over a coefficient field $\mathbb{K}$. A homogeneous word $w = (j_1, \ldots, j_d)$ represents $A_{j_1} \cdots A_{j_d}$. Up to degree $q$, we store a series as the sparse map

$$P(t) = 1 + \sum_{d=1}^{q} t^d \sum_{|w|=d} p_w w + O(t^{q+1}). \tag{14}$$

Multiplication is truncated convolution. An exponential stage is generated by

$$\exp(atA_j) = \sum_{k=0}^{q} \frac{(at)^k}{k!} A_j^k + O(t^{q+1}), \tag{15}$$

and all 31 stages are multiplied in their actual execution order. The coefficients $a$ are held as outward rational intervals. Thus every stored coefficient encloses the corresponding real algebraic coefficient, including the fourth-order Suzuki coefficient involving $4^{1/3}$.

Table 2: Compiler passes and the information they preserve.

| Pass | Conventional relaxation | Replacement | Verified invariant |
|---|---|---|---|
| Complete formula log | Bound stages or partial sums separately | Multiply all stages and take the formal log before norming | Truncated series identity and order conditions |
| Lie projection | Treat free words as unrelated monomials | Project homogeneous log defects into free-Lie elements | DSW identity at fixed degree |
| Concrete evaluation | Count generic nonzero commutators | Evaluate exact local Pauli products and phases | Symplectic multiplication and support |
| Translation quotient | Count translated clusters separately | Select a canonical orbit representative | Translation-equivalent configurations merge |
| Coefficient aggregation | Take absolute values by derivation path | Sum exact coefficients of identical Paulis | Operator represented by sparse ledger |
| Anticommuting grouping | Apply termwise Pauli-$\ell_1$ | Use exact group $\ell_2$ identities | Coverage and pairwise anticommutation |
| Finite-step closure | Use only leading order | Retain D4–D7 and rational tail | Global error upper bound |
| Adjacent integer search | Round a continuous estimate | Verify $r$ and $r-1$ exactly | Smallest integer accepted by this bound |

For $X = P - 1$, the formal logarithm is

$$\log P = \sum_{k=1}^{q} \frac{(-1)^{k+1}}{k} X^k + O(t^{q+1}). \tag{16}$$

No matrices and no floating-point eigensolvers enter this computation. The calculation is universal in the letters; only after the Lie defect is known do we substitute the concrete Heisenberg fragments.

### 5.2 Order, symmetry, and the full-stage cancellation budget

Write the ideal generator as $A = \sum_{j=1}^{4} A_j$ and the logarithm of a single formula step as

$$\log S_4(t) = tA + t^5 E_5 + t^7 E_7 + O(t^9). \tag{17}$$

Fourth order requires the degree-two through degree-four defects to vanish. Time symmetry, $S_4(-t) = S_4(t)^{-1}$, implies that the logarithm is odd, so its even homogeneous components vanish. The compiler checks these statements coefficient by coefficient in the word IR; it does not infer them merely from the name of the formula.

This global construction matters. If a bound is applied to each stage or to each recursive subformula before the product is assembled, opposite coefficients and Jacobi-related terms can no longer cancel. In our pipeline, absolute values first appear only after the complete formula has been projected, evaluated, translated, and aggregated. This ordering of operations is the main methodological distinction from a direct “expand–triangle-inequality” instantiation of a general commutator theorem.

### 5.3 Dynkin–Specht–Wever projection

The homogeneous component $L_d$ of a logarithm of group-like series is a Lie element. We use the Dynkin–Specht–Wever (DSW) map to expose that structure. For a word $w = x_1 \cdots x_d$, define

$$D(w) = [x_1, [x_2, \ldots, [x_{d-1}, x_d] \ldots]]. \tag{18}$$

The DSW theorem gives $D(L_d) = dL_d$ for homogeneous Lie elements [3]. Consequently,

$$E_d = L_d = \frac{1}{d} \sum_{|w|=d} \ell_w D(w), \tag{19}$$

where $\ell_w$ are the word coefficients obtained from equation (16). Equation (19) is used as a projection and as an independently checkable identity: expanding its right-hand side back into words must recover $L_d$ interval-wise.

**Proposition 5.1** (Semantic preservation of the Lie lowering pass)**.** *Assume the input interval map encloses every coefficient of the degree-d homogeneous formal logarithm and the DSW reconstruction check succeeds. Then the emitted nested-commutator interval map encloses the exact homogeneous Lie defect $E_d$.*

*Proof.* The DSW map is linear over the coefficient field. Interval scalar multiplication and addition are inclusion isotone, so applying $D/d$ to the input enclosures contains its application to the exact coefficients. The DSW identity fixes the exact image to $E_d$. The reconstruction check additionally guards against word-order, sign, and truncation errors in the implementation. □

### 5.4 Why this is an algorithmic improvement

The free-Lie stage is not a new product formula: it is a better compiler for a fixed formula. Its improvement is threefold. First, it preserves cross-stage cancellations. Second, it replaces derivation histories by a canonical operator expression. Third, it makes the later model substitution sparse: nested commutators that vanish under the local Pauli representation are removed exactly rather than charged by a worst-case commutator factor. The cost is an offline symbolic calculation whose size grows rapidly with degree. That cost is acceptable here because the certificate is generated once and verified much more cheaply than it is discovered.

## 6 Concrete Pauli Evaluation and Translation Quotienting

After free-Lie extraction, the compiler substitutes the four checkerboard fragments of the square-lattice Heisenberg Hamiltonian. This is the point at which generic commutator counting is replaced by exact model algebra.

### 6.1 Symplectic Pauli representation

Ignoring a separately stored phase, an $n$-qubit Pauli is represented by two bit strings $(x, z) \in \mathbb{F}_2^{2n}$ through the stabilizer/symplectic formalism [4], $P = i^\kappa X^x Z^z$. Multiplication is bitwise xor plus an exactly updated phase. Two Paulis commute exactly when their symplectic product vanishes:

$$[P(x, z), P(x', z')] = 0 \iff x \cdot z' + z \cdot x' = 0 \pmod 2. \tag{20}$$

Thus a nested commutator is evaluated without constructing a $2^{144} \times 2^{144}$ matrix. At each bracket, commuting pairs are deleted; anticommuting pairs produce the exact factor and phase. Duplicate bit strings are combined before the next bracket.

The local bond operator

$$h_{uv} = \frac{1}{4}(X_u X_v + Y_u Y_v + Z_u Z_v) \tag{21}$$

has only three Pauli summands. Each of the four fragments is a sum over a perfect matching of the periodic $12 \times 12$ lattice. Terms within a matching commute and their exponentials reduce to independent two-qubit bond propagators.

### 6.2 Locality before enumeration

The support of $[P, Q]$ is contained in $\mathrm{supp}(P) \cup \mathrm{supp}(Q)$, and the bracket vanishes when the strings commute. The implementation therefore carries connected bond clusters rather than global lattice objects. A degree-$d$ nested commutator can touch only a bounded neighborhood of its seed bond. Candidate bonds outside that neighborhood are never enumerated. This converts a nominally exponential global construction into a finite local calculation followed by a translation-density factor.

For the $L = 12$ benchmark the supports fit inside the injectivity radius of the torus for the degrees used in the explicit ledger. Periodic coordinates are nevertheless retained, and wraparound is checked rather than assumed away. This avoids silently importing an infinite-lattice result into a finite periodic instance.

### 6.3 Canonical translation representatives

Let $\tau_a$ translate a Pauli string by a lattice displacement $a$. We choose the lexicographically least serialization among all translates as the orbit representative

$$\mathrm{can}(P) = \min_{a \in \mathbb{Z}_L^2} \mathrm{serialize}(\tau_a P). \tag{22}$$

The coefficient map is then aggregated by this key. Canonicalization occurs before taking absolute values, so translated derivation paths that land on the same physical Pauli can cancel exactly.

**Proposition 6.1** (Correctness of translation aggregation)**.** *Let $Q = \sum_j c_j P_j$ be a translation-invariant operator written with a marked origin, and let $\bar{c}_P$ be the sum of all coefficients mapped to canonical representative $P$. On an $L \times L$ periodic lattice,*

$$Q = \sum_{a \in \mathbb{Z}_L^2} \tau_a \left( \sum_P \bar{c}_P P \right), \tag{23}$$

*with the conventional stabilizer multiplicities included in $\bar{c}_P$.*

*Proof.* Translation partitions the enumerated terms into disjoint group orbits. Within each orbit, linearity permits coefficient aggregation at a selected representative. Re-expanding the orbit sum recovers every translated term with the same multiplicity. The implementation records orbit and stabilizer counts, making the equality a finite combinatorial check. □

### 6.4 Exact aggregation and interval coefficients

Every coefficient is an interval $[c^-, c^+]$ with rational endpoints. When multiple derivation paths yield the same Pauli key, intervals are added first. Only then is an outward absolute upper bound formed:

$$|[c^-, c^+]|_\uparrow = \max\{|c^-|, |c^+|\}. \tag{24}$$

This order is essential. Replacing $\sum_j c_j$ by $\sum_j |c_j|$ earlier would discard the formula-specific and model-specific cancellations that the compiler was designed to retain.

The output of this pass is not merely a scalar norm. It is a complete sparse ledger of canonical Pauli keys and coefficient enclosures. That ledger can be hashed, independently replayed, regrouped by a future heuristic, or sampled for direct small-system matrix cross-checks. These properties turn a one-off symbolic calculation into reusable evidence.

# 7 Anticommuting Norm Certificates

After exact coefficient aggregation, the direct Pauli triangle inequality $\left\|\sum_j c_j P_j\right\| \le \sum_j |c_j|$ remains sound but unnecessarily loose. The leading right-generator defect contains many mutually anticommuting strings. Their norm is an $\ell_2$, not an $\ell_1$, quantity.

## 7.1 Weighted anticommuting groups

**Lemma 7.1** (Pairwise-anticommuting Pauli norm)**.** *Let $P_1, \ldots, P_k$ be Hermitian Pauli operators with $P_i^2 = I$ and $\{P_i, P_j\} = 0$ for $i \neq j$. For real coefficients $c_i$,*

$$\left\|\sum_{i=1}^{k} c_i P_i\right\| = \sqrt{\sum_{i=1}^{k} c_i^2}. \tag{25}$$

*Proof.* Squaring the sum gives $\sum_i c_i^2 I + \sum_{i<j} c_i c_j (P_i P_j + P_j P_i)$; all cross terms vanish. Hence every eigenvalue has magnitude $\sqrt{\sum_i c_i^2}$, proving the operator-norm identity. □

For interval coefficients we replace each $|c_i|$ by its outward rational upper endpoint and enclose the square root from above. If $\mathcal{G} = \{G_1, \ldots, G_s\}$ is a partition of the Pauli ledger into pairwise-anticommuting groups, the certified cell bound is

$$B_{\mathcal{G}} = \sum_{g=1}^{s} \left(\sqrt{\sum_{j \in G_g} |c_j|_{\uparrow}^2}\right)_{\uparrow}. \tag{26}$$

Singletons are permitted, so the method is never less sound than termwise norming and can be applied incrementally.

## 7.2 Grouping as a graph certificate

Construct the anticommutation graph whose vertices are canonical Pauli terms and whose edges join anticommuting pairs. A valid group is a clique; a valid certificate is a vertex-disjoint clique cover. Finding a minimum-weight cover is difficult and unnecessary for correctness. The discovery code uses a deterministic weighted greedy heuristic that prioritizes large coefficients and inserts a term only when it anticommutes with every current member.

The verifier does not trust the heuristic. For every emitted group it checks:

1. each referenced Pauli key exists in the coefficient ledger;
2. every key appears exactly once across the partition;
3. all distinct pairs satisfy equation (20) with symplectic parity one;
4. the squared-weight sum uses the declared coefficient enclosures; and

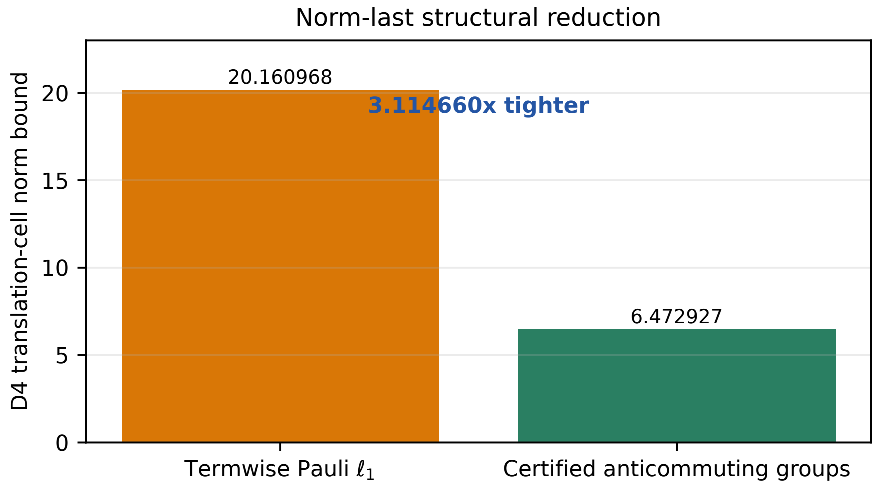


Figure 2: Leading D4 cell norm under direct Pauli-$\ell_1$ norming and the verified pairwise-anticommuting partition. Both values are computed after exact aggregation; the grouped bar is a rigorous upper bound.

5. the rational square-root endpoint squares to at least that sum.

These checks are polynomial in the output size and make optimality irrelevant to the trusted claim.

### 7.3 Leading-defect certificate

For the degree-four right-generator contribution D4, the concrete ledger has 75,324 canonical terms. The emitted partition contains 7,576 groups, with maximum group size ten. The resulting per-cell upper bound is

$$B_{D4}^{\mathrm{group}} = 6.472926505087888, \tag{27}$$

whereas direct post-aggregation Pauli-$\ell_1$ norming gives 20.160968407335066. The local norm reduction is therefore $3.114660484942633\times$. These are certificate statistics, not estimates from sampled matrices.

### 7.4 A local Heisenberg identity

The grouping strategy is reinforced by a small exact algebraic fact. Let $h = (XX + YY + ZZ)/4$ on two qubits and let $P$ be a two-qubit Pauli. Terms of $h$ that commute with $P$ do not contribute to $[h, P]$. The remaining Pauli products occur with exactly tracked phases and are pairwise anticommuting for the local cases used by the certificate. Consequently the commutator norm is obtained from a short Euclidean coefficient vector rather than by repeatedly using $\|[A, B]\| \le 2\,\|A\|\,\|B\|$.

This lemma is checked exhaustively over the 16 local Paulis. It supplies an independent local cross-check on the symplectic engine and explains why the concrete Heisenberg algebra is substantially smaller than a generic nested commutator count. It does not, by itself, prove the global bound; that role is played by the complete group membership file and finite-step ledger.

## 8 Finite-Step Right-Generator Certificate

An asymptotic statement $O(t^5)$ is insufficient at the target tolerance. The certified step count is obtained from a finite-$t$ right-generator bound, including explicit subleading terms and a rational tail.

### 8.1 From logarithmic defect to right generator

Let $S(t)$ denote one formula step and define its right generator

$$K(t) = S'(t)S(t)^{-1}. \tag{28}$$

For $Z(t) = \log S(t)$, differentiation of the exponential gives

$$K(t) = \mathrm{dexp}_{Z(t)}(Z'(t)) = \sum_{n\geq 0} \frac{1}{(n+1)!} \operatorname{ad}^n_{Z(t)} Z'(t). \tag{29}$$

Substituting equation (17) and collecting powers yields

$$K(t) - A = t^4 D_4 + t^5 D_5 + t^6 D_6 + t^7 D_7 + R_{\geq 8}(t), \tag{30}$$

where, writing $B = E_5$ and $C = E_7$, the exact coefficients through degree seven are

$$D_4 = 5B, \qquad D_5 = 2[A, B], \tag{31}$$

$$D_6 = 7C + \tfrac{2}{3}[A, [A, B]], \qquad D_7 = 3[A, C] + \tfrac{1}{6}[A, [A, [A, B]]]. \tag{32}$$

The implementation derives these coefficients from the truncated series rather than trusting the printed display. The identity clarifies why an exact leading logarithmic coefficient is not yet a global error certificate: commutators generated by the differential of the exponential contribute at the next powers.

### 8.2 Duhamel conversion to global error

Let $U(t) = e^{tA}$ in anti-Hermitian convention. Variation of constants gives the one-step estimate

$$\|S(\delta) - U(\delta)\| \leq \int_0^{\delta} \|K(s) - A\|\, ds. \tag{33}$$

Unitary telescoping over $r$ equal steps, with $\delta = T/r$, then yields

$$\|S(T/r)^r - U(T)\| \leq r \int_0^{T/r} \|K(s) - A\|\, ds. \tag{34}$$

Combining equations (30) and (34), a certificate with norm upper bounds $b_j \geq \|D_j\|$ and a nonnegative tail majorant $\rho(s)$ establishes

$$\mathcal{E}(r) = r \left( \sum_{j=4}^{7} \frac{b_j}{j+1} \left(\frac{T}{r}\right)^{j+1} + \int_0^{T/r} \rho(s)\, ds \right). \tag{35}$$

All quantities in the machine certificate are rationals. Decimal values in this paper are outward-rounded summaries only.

### 8.3 Rational tail closure

The exact ledger is stopped after D7. Higher degrees are bounded by a geometric majorant derived from the absolute stage coefficients and local commutator growth. In abstract form, if the degree-$k$ coefficient is at most $Mq^{k-k_0}$ for $0 \leq qs < 1$, then

$$\rho(s) \leq \frac{Ms^{k_0}}{1 - qs}. \tag{36}$$

The verifier checks the rational preconditions, including positivity of the denominator on $[0, T/r]$, and integrates an outward upper enclosure. This is deliberately more conservative than extrapolating the observed low-degree coefficients. It ensures that no uncomputed term is silently discarded.

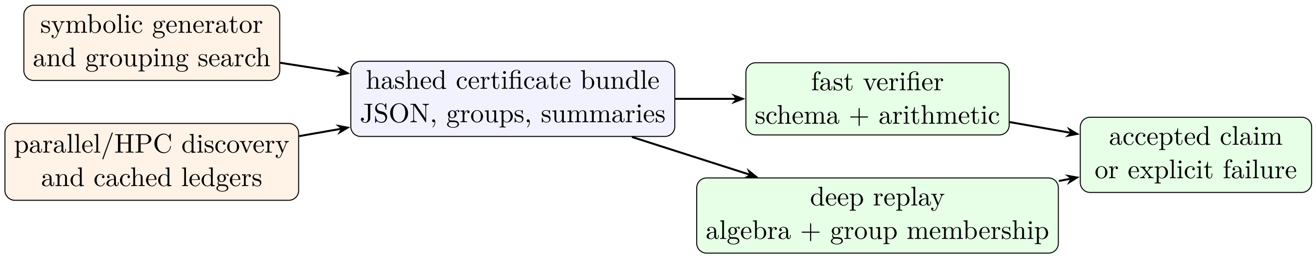


Figure 3: Trust boundary. Parallel discovery and heuristic grouping are untrusted producers. The accepted result depends on explicit evidence and two verification paths.

**Theorem 8.1** (Soundness of an accepted finite-step certificate)**.** *Suppose the verifier accepts (i) interval enclosures for D4–D7, (ii) a valid operator-norm certificate for each declared coefficient, (iii) a tail majorant satisfying equation (36) on* $[0, T/r]$*, and (iv) exact rational evaluation* $\mathcal{E}(r) \le \varepsilon$*. Then*

$$\left\| S(T/r)^r - e^{TA} \right\| \le \varepsilon. \tag{37}$$

*Proof.* The accepted coefficient and tail checks bound $\|K(s) - A\|$ term by term on the complete integration interval. Integrating preserves the inequality. Equation (34) then bounds the global error, and the exact rational comparison with $\varepsilon$ completes the proof. □

### 8.4 Adjacent-step minimality under the certified bound

The search evaluates integers, not a rounded real-valued estimate. It returns the least $r$ accepted by equation (35) and emits the rejected neighbor. For this benchmark,

$$\mathcal{E}(97) \le 9.958938494314325 \times 10^{-7} < 10^{-6}, \tag{38}$$
$$\mathcal{E}(96) \le 1.050565873970784 \times 10^{-6} > 10^{-6}. \tag{39}$$

The second inequality says that 96 is not certified by *this bound*; it does not prove that the physical simulation error at 96 steps exceeds the tolerance. This distinction is retained throughout the paper.

## 9 Certificate Architecture and Independent Verification

Large symbolic calculations are difficult to trust by inspection. We use a proof-carrying architecture: discovery may be complex, but the final claim is accepted only after a narrower verifier rechecks explicit evidence.

### 9.1 Trusted and untrusted components

The trusted claim comprises the mathematical lemmas in this paper, the small verification programs, exact integer/rational arithmetic, and the cryptographic binding between manifest and sidecars. It does not require trusting the job scheduler, the heuristic's optimality, cached stdout, or a claimed successful HPC run.

### 9.2 Certificate contents

The schema-v3 certificate records:

- the complete benchmark tuple, normalization, formula, time, tolerance, and primary resource metric;
- the pinned theorem instantiation used for the baseline;
- coefficient precision policies and rational interval endpoints;
- D4 group-certificate path, SHA-256 digest, term count, group count, maximum size, and norm enclosure;
- D4–D7 and tail contributions at the accepted integer;
- the accepted and previous-step global-error rationals;
- integer resource arithmetic and the exact improvement ratio; and
- auxiliary audit results explicitly separated from the primary claim.

Large membership data are sidecars rather than embedded in prose. The manifest hash prevents a reviewer from inadvertently validating a different file with the same name.

### 9.3 Fast and deep modes

Fast verification checks the schema, paths, hashes, integer identities, rational comparisons, resource formulas, and headline invariants. It is intended for continuous integration and reviewer orientation. Deep verification additionally replays coefficient aggregation, checks every group for exact coverage and pairwise anticommutation, verifies all rational square roots, and reproduces the finite-step sum.

The two modes serve different purposes. Fast verification answers whether a checked-in bundle is internally bound and arithmetically self-consistent. Deep verification answers whether the large algebraic witness supports its declared norm. Neither mode treats a rounded decimal in a PDF as evidence.

### 9.4 Negative tests and failure localization

A certificate system is useful only if nearby corruptions fail. The test suite mutates representative fields: a group member, a coefficient endpoint, the accepted step count, a resource total, and a sidecar digest. Each mutation must cause a nonzero verifier result. Further unit tests cover Pauli phase multiplication, symplectic commutation, periodic canonicalization, rational square-root enclosure, and the local 16-Pauli lemma.

Failures are reported by layer. A hash mismatch is distinguished from a schema error; a duplicate group member from a non-anticommuting pair; a failed tail precondition from an error-bound comparison. This makes independent reproduction tractable: a reviewer need not rerun the entire discovery pipeline to diagnose a damaged artifact.

### 9.5 Claim classes

We label statements by evidence class:

**Certified.** Derived from a manifest-bound witness and accepted by the verifier, such as the 97-step result and integer resource counts.

**Cross-checked.** Supported by a structurally independent small-system or exhaustive local calculation, but not the main large-system proof.

**Conditional.** An arithmetic consequence under a stated future premise, such as the norm budget required for a fivefold result.

Table 3: Certified resource comparison under the common compilation model.

| Method | Steps | Merged groups | Bond propagators | CNOT upper |
|---|---|---|---|---|
| Pinned published-theorem control | 393 | 11,791 | 848,952 | 2,546,856 |
| Proof-carrying model-aware bound | 97 | 2,911 | 209,592 | 628,776 |
| Candidate/control | 0.2468 | 0.2469 | 0.2469 | 0.2469 |

**Prospective.** A research direction or engineering optimization with no claimed current performance.

This vocabulary prevents an observed trend, a possible optimization, and a completed theorem from being conflated in the abstract or conclusion.

# 10 Certified Results

This section reports only values bound to the checked-in certificate. The primary metric is the compiled CNOT upper bound; merged fragment exponentials and bond propagators are reported because they make the integer cost model transparent.

## 10.1 Headline resource reduction

The candidate certificate accepts $r = 97$ steps. Adjacent identical fragment exponentials merge across recursive boundaries, so a 31-stage step sequence contains

$$G(r) = 30r + 1 \tag{40}$$

merged group exponentials. Each group is one perfect matching with 72 bonds. The selected two-qubit bond-propagator compilation is charged at three CNOTs. Therefore

$$B(r) = 72G(r), \qquad C(r) = 3B(r) = 216G(r). \tag{41}$$

Because the last three resource measures are proportional under equation (41), their exact improvement factor is

$$\frac{11{,}791}{2{,}911} = 4.050498110614909\ldots. \tag{42}$$

Equivalently, 8,880 of 11,791 merged groups are removed, a reduction of 75.31167839877873%. The claimed multiplier is an exact rational comparison; the decimal is for readability.

## 10.2 Finite-step error ledger

At 97 steps the global upper bound is the sum of five outward rational contributions shown in table 4. The leading D4 term is only about half the final value; D5–D7 and the tail are quantitatively material. This demonstrates why a leading-order-only prediction would not be an adequate certificate at the boundary.

The adjacent-step evidence is decisive for the integer search: 97 is accepted at $9.958938494314325 \times 10^{-7}$, whereas the same bound at 96 evaluates to $1.050565873970784 \times 10^{-6}$. The margin at 97 is about $4.11 \times 10^{-9}$, so outward arithmetic and tail closure are not cosmetic; ordinary double-precision summaries are not the source of the comparison.

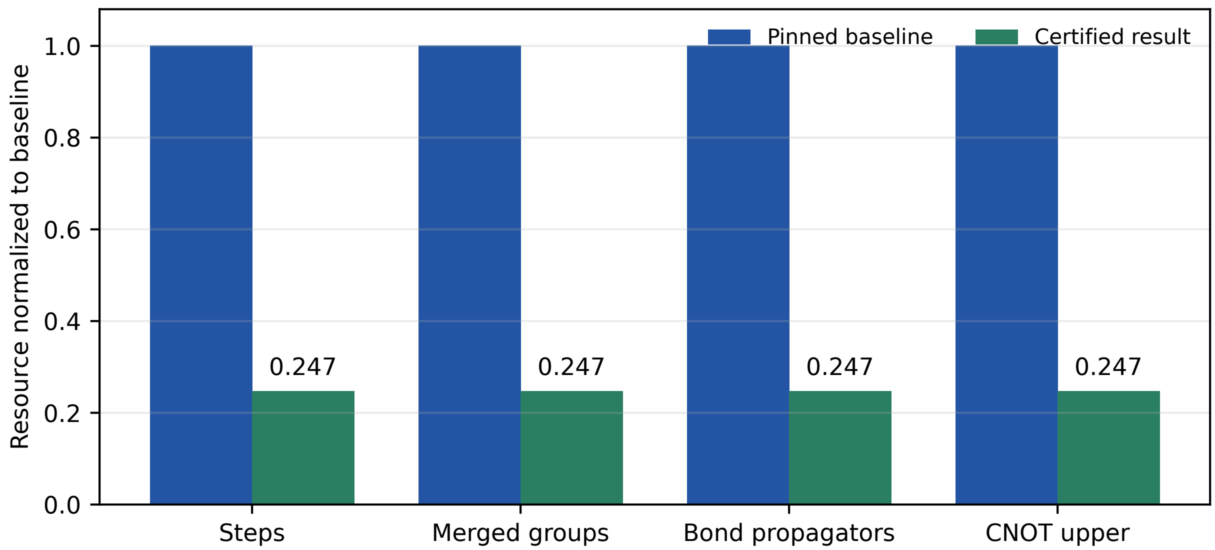


Figure 4: Resources for the pinned theorem instantiation and the certified model-aware result. Values are exact integers; axes are scaled separately to keep the three compiled metrics readable.

Table 4: Outward decimal summaries of exact rational contributions at $r = 97$.

| Contribution | Global error upper | Fraction of total (approx.) |
|---|---|---|
| D4 | $5.264367936822258 \times 10^{-7}$ | 52.86% |
| D5 | $1.383487536744612 \times 10^{-7}$ | 13.89% |
| D6 | $1.579057000594836 \times 10^{-7}$ | 15.86% |
| D7 | $1.894657080691470 \times 10^{-8}$ | 1.90% |
| Tail | $1.542560312083473 \times 10^{-7}$ | 15.49% |
| Total | $9.958938494314325 \times 10^{-7}$ | 100% |

### 10.3 Algebraic compression statistics

The D4 group sidecar contains 75,324 coefficient-bearing Pauli terms divided into 7,576 verified groups. The largest group has ten members. The grouped translation-cell norm 6.472926505087888 replaces a post-aggregation direct Pauli-$\ell_1$ value 20.160968407335066, giving the local factor 3.114660484942633. This local factor and the final resource factor need not coincide: the latter also reflects the nonlinear inversion of the step bound, the subleading terms, the tail, and the integer merge rule.

An auxiliary exact D5 computation produces 605,832 terms, 123,106 groups, and a per-site upper bound 11.23706750025697. It is retained as evidence for a future tightening effort. It does not replace the complete D4–D7 global ledger and is not used to claim a lower accepted step count.

### 10.4 Small-system structural cross-check

On a degenerate periodic $2 \times 2$ instance at the same 97-step count, direct dense evaluation gives operator-norm error $9.128277711816321 \times 10^{-10}$. The schema-v3 certificate specialization gives the upper bound $2.766666666666666 \times 10^{-8}$. The latter dominates the measured error and remains below $10^{-6}$. This calculation checks stage order, normalization, periodic bookkeeping, and the direction of the inequality using an independent dense route. It is not an empirical proof of the $12 \times 12$ result.

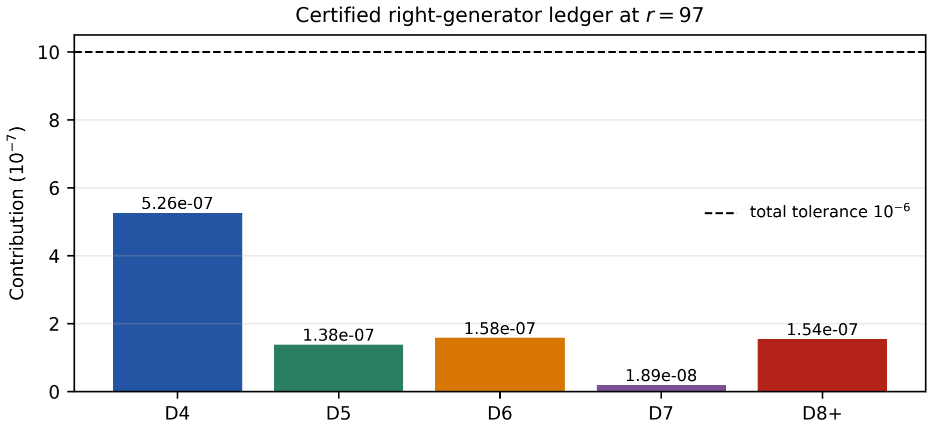


Figure 5: Finite-step error ledger at the accepted integer. The dashed line marks the total tolerance, not an independent allowance for each term.

Table 5: Auxiliary recursive-order audit stored in the authoritative main certificate.

| Order | Steps | Stages/step | Merged groups | Error upper summary |
|---|---|---|---|---|
| 2 | 29,964 | 7 | 179,785 | $9.9999203748666826 \times 10^{-7}$ |
| 4 | 808 | 31 | 24,241 | $9.9682621337975879 \times 10^{-7}$ |
| 6 | 370 | 151 | 55,501 | $9.8463753882538689 \times 10^{-7}$ |
| 8 | 335 | 751 | 251,251 | $9.9829085642259163 \times 10^{-7}$ |

### 10.5 Recursive Suzuki audit

For context, the main certificate records a separate audit of standard recursive Suzuki orders under its declared audit assumptions. The values in table 5 are taken from the manifest-bound schema-v3 JSON; they are not used in the primary 4.05× comparison.

This audit illustrates that increasing formal order does not monotonically reduce compiled cost at a fixed precision: the recursive stage multiplier can outgrow the reduction in steps. It should not be confused with a literature- wide optimal-formula search.

## 11 Related Work and Positioning

The relevant literature has diversified beyond a single sequence of tighter versions of the same bound. Some works strengthen worst-case product-formula analysis; others restrict the state or error metric; still others change the simulation algorithm. A meaningful position therefore requires a taxonomy rather than a one-dimensional “SOTA” label.

### 11.1 Direct product-formula bounds

Nearly optimal lattice simulation by high-order product formulas established the favorable scaling of local formulas for geometrically local Hamiltonians [5]. The commutator-scaling framework of Childs et al. then provided general explicit error representations in terms of nested commutators [1]. Model-specific evaluation has also been developed for the Fermi–Hubbard model [6], while strong-error analyses clarify the distinction between vector and operator convergence for unbounded or chemistry-motivated settings [7]. Lower-bound work studies when cancellation cannot make the Trotter error arbitrarily smaller than upper estimates [8].

Our work remains in this direct, deterministic, operator-norm branch. It does not introduce a new asymptotic theorem. Its contribution is an automated finite-instance sharpening layer after the general theorem has identified the relevant commutator structure. The proof object retains concrete coefficients and Pauli relations that a broad theorem cannot economically tabulate.

### 11.2 Restricted-state and average-case guarantees

Low-energy-subspace simulation can replace global energy scales by effective low-energy quantities when the initial state is promised to lie in an appropriate sector [9]. Average-case analysis can also yield speedups under distributional or norm-averaged criteria [10]. These guarantees may be more relevant for a given application than a worst-case operator norm, but they answer a different question. The present certificate makes no state promise and therefore remains uniform over all input states.

### 11.3 Changing the approximation family

Multi-product formulas combine several product-formula circuits to cancel leading errors. Recent work supplies explicit commutator scaling, dynamic coefficient selection, and time-dependent extensions [11–14]. Chebyshev interpolation and randomized order-doubling similarly alter how approximants are combined [15, 16]. Time-dependent product-formula constructions such as THRIFT exploit an interaction-picture decomposition [17]. LCU-based Trotter compensation and error-mitigation approaches add coherent or classical correction layers [18, 19]; more recent nested- commutator compensation seeks polylogarithmic precision without ancillas [20].

These methods can asymptotically or practically outperform repetition of a single deterministic Suzuki step. Their costs include extra circuit variants, sampling, negative or complex weights, success amplification, ancillas, or different depth/gate accounting. Our certified result keeps the original formula family fixed and reduces only the justified repetition count. It is therefore complementary: the same proof-carrying norm compiler could in principle supply coefficient bounds to an extrapolation or compensation method, but such an extension is not established here.

### 11.4 Formula and ordering optimization

Optimized Trotter–Suzuki coefficient sets can lower the stage count or error constant relative to recursive constructions [21]. Commutation-aware ordering studies show that the placement of Heisenberg-type terms can materially change realized error [22]. Practical error estimators trade some formal generality for computable instance-level predictions [23]. Those directions optimize the source formula or estimate its behavior. Here the source stage sequence is fixed and the novelty is in rigorously compiling its complete defect into a checkable upper bound.

### 11.5 Position of the present result

Within the narrow target class, this work establishes a strong finite-instance result: a $4.050498\times$ certified reduction over a pinned application of a published general theorem, while preserving the formula, operator-norm metric, and gate-accounting rule. Its scientific role is best described as a *model-aware proof compiler* between general theory and executable resource estimation.

It would be inaccurate to call the result an unconditional global state of the art in Hamiltonian simulation. More recent algorithms can have superior precision scaling or lower costs under changed assumptions. Conversely, their headline numbers do not invalidate this result because they are not necessarily certificates for the same fixed deterministic circuit. The paper's defensible novelty is the

Table 6: Methodological comparison. "Same target" means a deterministic, uniform operator-norm certificate for the fixed formula and compilation model of section 2.

| Line of work | Main mechanism | Guarantee/cost change | Relation to our method | Same target? |
|---|---|---|---|---|
| Commutator scaling [1] | General nested-commutator error representation | Worst-case direct product formula | Supplies the pinned control and theoretical foundation | Yes, at theorem level |
| Model-specific Hubbard bounds [6] | Evaluate locality and commutators for another model | Model-aware direct bound | Similar model-specific philosophy; different Hamiltonian and certificate machinery | Partly |
| Low-energy simulation [9] | Restrict input energy sector | State-dependent promise | Could be combined with concrete algebra; not used here | No |
| Average-case speedup [10] | Weaken worst-case metric to an average | Distribution-dependent error | Orthogonal norm objective | No |
| Dynamic/explicit MPF [11–13] | Linear combination of several formula lengths | New coefficients, sampling or coherent combination | Our defect bounds may inform coefficients; current circuit family differs | No |
| Interpolation / randomization [15, 16] | Combine runs or random orderings | Changed estimator/algorithm | Potential outer layer around certified formulas | No |
| THRIFT [17] | Time-dependent interaction-picture formula | Changed decomposition and circuit | Could benefit from proof-carrying instance analysis | No |
| LCU/mitigation [18–20] | Correct or cancel Trotter error | Ancilla, sampling, or compensation overhead | Uses error structure actively; ours only reduces safe repetitions | No |
| Formula/ordering optimization [21, 22] | Change stages, coefficients, or ordering | Different source circuit | Complementary front end to our compiler | No |
| This work | Complete-formula Lie extraction, exact Pauli aggregation, verified anticommuting norms | Same formula; smaller certified repetition count | Finite-instance proof-carrying sharpening | Yes |

combination of full-formula symbolic cancellation, concrete Pauli norm compression, nonasymptotic closure, and an independently checkable artifact.

# 12 Limitations and Research Roadmap

The present evidence supports a precise but deliberately narrow conclusion. This section identifies the boundaries and shows how each could be addressed without weakening the current claim.

## 12.1 Finite instance rather than universal scaling

The numerical certificate is for the periodic $12 \times 12$ isotropic Heisenberg benchmark at $T = 1$ and $\varepsilon = 10^{-6}$. Locality suggests that many symbolic densities extend to larger tori beyond the support radius, but the current manifest does not claim a theorem uniform in $L$, anisotropy, time, or tolerance. A scaling paper would require parameterized wraparound proofs, analytic dependence of every ledger term, and certificates across a grid of $(L, T, \varepsilon)$ values.

## 12.2 Fixed product formula and cost model

The candidate and control use the same five-copy fourth-order Suzuki formula. No search over modern optimized splittings is included. Similarly, the three-CNOT bond cost is an upper-bound compilation model, not a full fault-tolerant resource estimate. Native two-qubit gates, routing, parallel depth, synthesis precision, error correction, and logical-state preparation can change the operational optimum. Future work should make the cost model a pluggable certificate field and report Pareto fronts for group count, two- qubit gates, and depth.

## 12.3 Certificate size and discovery cost

The exact D4 and D5 ledgers are large. They are appropriate for an offline proof artifact but not yet an interactive compiler pass. The deep verifier is much simpler than the generator but still scales with every listed term and pair checked within a group. Merkleized chunks, streaming verification, symmetry-orbit proofs, and formally verified local rewrite kernels could reduce storage and trust without replacing explicit evidence by an opaque claim.

## 12.4 The fivefold boundary is open

A fivefold resource ratio would be obtained at $r = 78$ because

$$\frac{11{,}791}{30 \cdot 78 + 1} = \frac{11{,}791}{2{,}341} = 5.036736437419906\ldots. \tag{43}$$

This is arithmetic, not a 78-step error certificate. The checked artifacts identify three unresolved proof gates: a tighter translation-coupled D4 norm at the 78-step budget, explicit D8 information with a delayed high-degree tail, and a complete exact global ledger at $r = 78$. Exact D5 grouping alone does not discharge these obligations.

A disciplined optimization sequence is therefore: (1) profile the current 97-step error ledger; (2) improve the dominant D4 norm while preserving a verifiable partition; (3) emit D8 explicitly; (4) move the geometric tail to a higher starting degree; (5) regenerate exact contributions at 78; and (6) accept the point only if the rational sum is at most $10^{-6}$. GPU or HPC parallelism can accelerate term generation and graph search, but cannot replace the last verifier step.

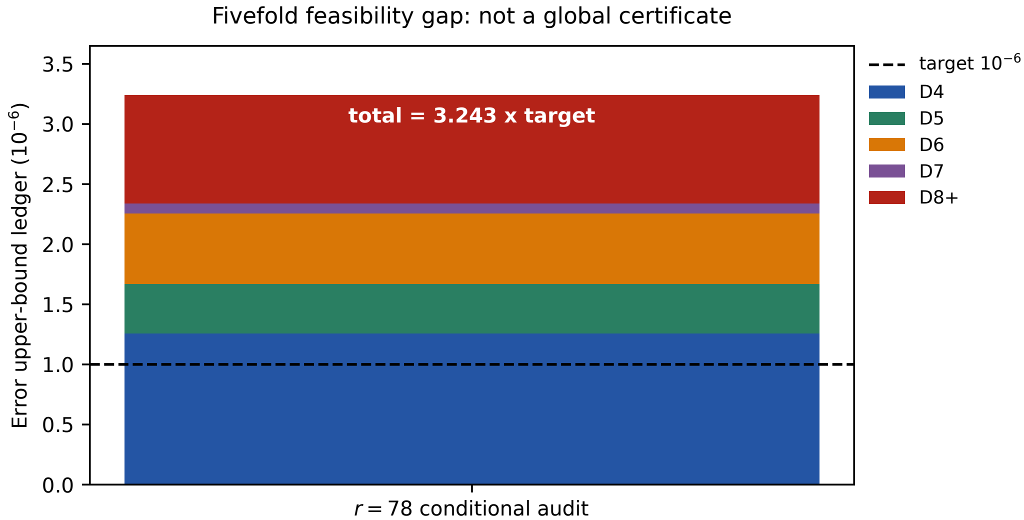


Figure 6: Conditional error ledger at the fivefold resource threshold. The stacked value exceeds the target; $r = 78$ is therefore not certified. The 97-step result remains the lowest accepted point of the current bound.

### 12.5 Optimality is relative to the bound

The rejection at 96 establishes adjacent minimality only under the emitted upper bound. It is not a lower bound on the true operator error. Direct large-system matrix simulation is infeasible, and lower-bound techniques have not been specialized here to close the gap. Consequently the candidate may still be physically accurate with fewer steps. The scientific claim concerns what can be guaranteed, not an assertion that 97 executions are necessary.

### 12.6 Toward broader validation

The most valuable next experiments are not additional decimal precision at the current point. They are adversarial reproductions: an independent Pauli engine, certificates for open and periodic boundaries, anisotropic XXZ couplings, alternate fragment orderings, and comparisons against optimized splittings. These tests would reveal whether the dominant gains arise from universal compiler structure or accidental symmetries of the isotropic benchmark.

## 13 Reproducibility and Artifact Protocol

The repository is organized so that a reader can first validate the headline claim in seconds and then choose progressively more expensive checks. No private cluster is required to verify the submitted certificate.

### 13.1 Environment and commands

The reference implementation requires Python 3.11 or later and declares NumPy, SciPy, and SymPy dependencies. From the `issue128` directory, the standard sequence is:

```
python -m pip install -e '.[test]'
pytest -q
PYTHONPATH=src python scripts/verify.py \
```

```
  certificates/issue128-certificate.json
PYTHONPATH=src python scripts/build_d5_certificate.py --verify-only
PYTHONPATH=src python scripts/package_delivery.py --check
shasum -a 256 -c artifacts/SHA256SUMS
```

The expensive algebraic replay is requested explicitly:

```
PYTHONPATH=src python scripts/verify.py \
  certificates/issue128-certificate.json --deep
```

The main certificate can be regenerated with

```
PYTHONPATH=src python scripts/build_v3_certificate.py
```

Long research-reproduction tests are marked `slow` and excluded by the default Pytest configuration. This separation keeps ordinary continuous integration responsive without disguising the cost of complete regeneration.

### 13.2 Claim-to-evidence map

Table 7: Where each principal claim is established and how it is checked.

| Claim | Primary evidence | Independent check |
|---|---|---|
| 393-step pinned control | Baseline rational and center in `issue128-certificate.json` | Deep regeneration of all 31 centers |
| 97 accepted, 96 rejected | Exact global and previous-step rational pairs in the main certificate | Exact comparison against $1/10^6$ |
| 75,324 D4 terms and 7,576 groups | Hashed `issue128-d4-groups.json` | Coverage, symplectic pair checks, and rational square-root replay |
| D4 bound 6.472926505087888 | Group membership and coefficient endpoints | Verifier squares each upper root and resums |
| 11,791 and 2,911 groups | Step counts plus $G(r) = 30r + 1$ | Direct integer identity checks |
| Bond and CNOT totals | Cost-model fields and claimed-resource block | $72G(r)$ and $216G(r)$ recomputation |
| 4.050498× improvement | Exact pair $(11791, 2911)$ | Fraction comparison, not rounded decimal |
| Small-system consistency | Dense $2 \times 2$ artifact | Independent matrix exponential and operator norm |
| D5 follow-up only | Hashed compressed D5 sidecar and summary status | `-verify-only`; explicit `not_certified` fivefold status |

### 13.3 Immutable bindings

At the frozen delivery point the primary digests are

$$\begin{aligned}\text{SHA256(main certificate)} &= \texttt{0a09623ce3b292a3637065c870fb3153b}\ldots,\\ \text{SHA256(D4 sidecar)} &= \texttt{a397414bb0229fb1ebdb38798aa781fb8}\ldots,\\ \text{SHA256(D5 sidecar)} &= \texttt{c5e8968a93b4497b41fe42c0e364324}\ldots.\end{aligned}$$

The complete 64-hex-character values are stored in the package manifest and summary. A printed prefix is for human orientation only; verification uses the complete digest.

The manuscript is a derived view and is not part of the frozen proof bundle. If the manuscript, figures, or wording changes, the mathematical claim remains bound to the same JSON and sidecars. Conversely, any change to a certificate requires regeneration of its digest and all summaries that cite it.

### 13.4 Machine-readable and human-readable outputs

The delivery contains a main certificate JSON, large exact sidecars, a compact JSON summary, a plain-text summary, verification transcript, checksums, source code, tests, a short technical report, and this long-form manuscript. The machine-readable objects are authoritative for exact numerators and denominators. Human documents explain the method and use outward decimal summaries; they should never be parsed as the proof source.

### 13.5 Research integrity statement

Software-assisted symbolic algebra, exact arithmetic, automated tests, and language-model-assisted drafting were used in preparing the artifact and manuscript. Scientific responsibility remains with the author. Every headline numerical statement is checked against the frozen certificate by a separate manuscript validation script; literature statements are cited to their primary publications. No generative system output is accepted as a mathematical witness without executable or deductive verification.

## 14 Conclusion

We have presented a proof-carrying, model-aware compiler for rigorous product-formula resource bounds. Its central design rule is simple: preserve algebraic structure until all formula-specific, model-specific, translational, and anticommuting relations have been exposed; apply irreversible norm inequalities only afterward; and emit enough evidence for an independent program to recheck the result.

For the periodic $12 \times 12$ isotropic Heisenberg benchmark, this procedure certifies the original five-copy fourth-order Suzuki circuit at 97 steps. The pinned general-theorem instantiation requires 393 steps under the common comparison protocol. After deterministic stage merging and bond compilation, the corresponding group, bond, and CNOT resources fall by the exact factor $11791/2911 = 4.050498110614909\ldots$. The result includes the rejected 96-step neighbor, D4–D7 contributions, a rational high-degree tail, full D4 anticommuting membership, resource arithmetic, and hash bindings.

The broader lesson is not that one Hamiltonian has exhausted the possibilities of Trotter simulation. It is that a general error theorem and a concrete gate estimate should be connected by a transparent compiler rather than a chain of unrecorded inequalities. The current artifact occupies that middle layer. A fivefold threshold remains a well-specified research target, but it will be claimed only after the missing D4, D8, and tail obligations close in the same machine-checkable form.

## A Right-Generator Expansion Details

This appendix records the finite-order identity used by the certificate. Let

$$Z(t) = \log S_4(t) = tA + t^5 B + t^7 C + O(t^9). \tag{44}$$

Using

$$\mathrm{dexp}_Z(Z') = Z' + \frac{1}{2}[Z, Z'] + \frac{1}{6}[Z, [Z, Z']] + \frac{1}{24}[Z, [Z, [Z, Z']]] + \cdots, \tag{45}$$

and collecting powers through $t^7$ gives

$$S_4'(t)S_4(t)^{-1} - A = 5t^4B + 2t^5[A,B] + t^6\left(7C + \frac{2}{3}[A,[A,B]]\right) + t^7\left(3[A,C] + \frac{1}{6}[A,[A,[A,B]]]\right) + R_{\geq 8}(t). \tag{46}$$

For example, the $t^5[A,B]$ coefficient comes from $\frac{1}{2}[tA, 5t^4B] + \frac{1}{2}[t^5B, A] = 2t^5[A,B]$. The later coefficients follow by the same homogeneous bookkeeping. The implementation constructs the series in the word algebra and regression-tests equation (46); it does not substitute the printed identity as an unchecked constant table.

The global contribution of a monomial $t^jD_j$ follows from equation (34):

$$r\int_0^{T/r} s^j\,\|D_j\|\,ds = \frac{T^{j+1}\,\|D_j\|}{(j+1)r^j}. \tag{47}$$

This explains the degree labels in the result ledger and provides a direct dimensional check on the inverse powers of $r$.

# B Local Heisenberg Pauli Audit

The single-bond lemma can be checked without symbolic software. Let $Q_1 = XX$, $Q_2 = YY$, and $Q_3 = ZZ$. For a phase-free two-qubit Pauli $P = P_1 \otimes P_2$, the number of $Q_j$ that anticommute with $P$ is either zero or two. One compact enumeration is shown in table 8; rows and columns give the one-qubit factors.

Table 8: Number of anticommuting terms among $XX, YY, ZZ$ for every local two-qubit Pauli.

| $P_1 \backslash P_2$ | $I$ | $X$ | $Y$ | $Z$ |
|---|---|---|---|---|
| $I$ | 0 | 2 | 2 | 2 |
| $X$ | 2 | 0 | 2 | 2 |
| $Y$ | 2 | 2 | 0 | 2 |
| $Z$ | 2 | 2 | 2 | 0 |

If the count is zero, $[h,P]=0$. If it is two, each contributing Hamiltonian coefficient has magnitude 1/4 and the Pauli commutator supplies magnitude two, giving two output coefficients of magnitude 1/2. Their Pauli-$\ell_1$ sum is one. Hence

$$\|[h,P]\|_{1,\mathrm{P}} \leq 1 = \|P\|_{1,\mathrm{P}} \tag{48}$$

for a unit Pauli input, and linearity extends the stated growth majorant. The generic bound obtained by summing the three Hamiltonian Paulis independently would give 3/2; the exact parity enumeration removes the impossible third contribution.

# C Verifier Pseudocode

Deep mode inserts regeneration of the baseline center scan, complete formula logarithm, DSW defects, Pauli coefficient map, and higher-degree majorants before the certificate comparisons. Importantly, both modes fail closed: unknown schema versions, absent files, extra group members, and invalid interval orientation are errors rather than warnings.

**Algorithm 2** Independent verification of the principal certificate

**Require:** main certificate $C$ and sidecar directory

validate schema version and required benchmark fields<br>
recompute SHA-256 for every referenced sidecar<br>
verify formula, normalization, $T$, $\varepsilon$, and cost-model identities<br>
load the D4 coefficient ledger and declared group partition<br>
verify exact coverage: no missing, duplicate, or unknown Pauli key<br>
**for** every group $G$ **do**<br>
    **for** every unordered pair $(P, Q) \subseteq G$ **do**<br>
        require symplectic parity$(P, Q) = 1$<br>
    **end for**<br>
    recompute the rational squared-weight sum<br>
    require declared upper root squared $\geq$ recomputed sum<br>
**end for**<br>
sum all group roots and compare to the D4 cell enclosure<br>
verify D5–D7 and tail preconditions and contributions<br>
sum the accepted-step global rational; require result $\leq \varepsilon$<br>
sum the previous-step rational; require result $> \varepsilon$<br>
recompute $G(r)$, bond propagators, CNOT totals, and exact ratio<br>
return a structured success record; fail closed on any mismatch

# D Certificate Schema Sketch

Table 9: Selected schema-v3 fields and verification obligations.

| Field | Type | Obligation |
|---|---|---|
| `benchmark` | object | Exact model, normalization, size, time, tolerance, and primary metric match the claim |
| `published_baseline.site_density_upper` | integer pair | Denominator is positive and deep regeneration is enclosed |
| `candidate.d4_certificate.path` | relative path | Resolves inside the bundle; no path traversal |
| `candidate.d4_certificate.sha256` | 64-hex digest | Equals the bytes of the resolved sidecar |
| `candidate.contributions` | rational pairs | Each pair has positive denominator and agrees with recomputed bounds |
| `candidate.global_error_upper` | rational pair | Equals or dominates the contribution sum and is below tolerance |
| `candidate.previous_step_error_upper` | rational pair | Correctly evaluated at $r-1$ and exceeds tolerance |
| `cost_model` | integers/rule | Resource identities are evaluated without floating point |
| `claimed_resources` | integer object | Every total is reconstructed from the step count and cost model |
| `claims.exact_improvement_ratio` | integer pair | Equals the declared baseline and candidate group counts |

# E Additional Interpretation of the Improvement

Several ratios appear in the analysis and should not be conflated:

$$\text{D4 norm tightening} = \frac{20.160968407335066}{6.472926505087888} = 3.114660484942633\ldots, \tag{49}$$

$$\text{step ratio} = \frac{393}{97} = 4.051546391752577\ldots, \tag{50}$$

$$\text{compiled resource ratio} = \frac{30 \cdot 393 + 1}{30 \cdot 97 + 1} = \frac{11791}{2911} = 4.050498110614909\ldots. \tag{51}$$

The step and resource ratios differ because the first and last fragment groups merge to give the additive one in equation (40). The norm ratio is not expected to equal either: a fourth-order global error scales roughly as $r^{-4}$ only in a leading asymptotic regime, while the actual accepted point contains D5–D7, a tail, and integer rounding.

The improvement relative to the challenge's twofold target is

$$\frac{(11791/2911)}{2} = 2.025249055307454\ldots. \tag{52}$$

This means the achieved multiplier is just over twice the requested multiplier; it does not mean an eightfold resource reduction.

# F Checklist for Independent Review

A reviewer can audit the result in increasing order of cost:

1. Read the benchmark tuple and confirm that candidate and control use the same Hamiltonian, formula, tolerance, and cost model.
2. Run the manuscript headline validator and the package checksum check.
3. Run the fast verifier and inspect its structured output.
4. Confirm the exact integer map from 393/97 steps to all three resources.
5. Run the default unit suite, including mutation and local-Pauli tests.
6. Run D5 `-verify-only` to confirm the follow-up is separate and marked noncertified at 78 steps.
7. If computational resources permit, run the deep verifier to regenerate the formula-specific algebra and baseline center scan.
8. Rebuild this PDF and visually inspect figures, tables, citations, and page boundaries; PDF appearance is a communication check, not a proof check.

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